

BMJ Open Physical activity levels, recreational screen time, sleep quality and mood among young adult healthcare students at an international university in Bahrain: a cross-sectional study

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To cite: AlKhenazi AK, Shakeeb FN, Fredericks S, *et al*. Physical activity levels, recreational screen time, sleep quality and mood among young adult healthcare students at an international university in Bahrain: a cross-sectional study. *BMJ Open* 2025;**15**:e093655. doi:10.1136/bmjopen-2024-093655

► Prepublication history for this paper is available online. To view these files, please visit the journal online (<https://doi.org/10.1136/bmjopen-2024-093655>).

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Received 12 September 2024
Accepted 25 April 2025



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ABSTRACT

Objectives To investigate levels of recreational physical activity, screen time, sleep quality and mood in undergraduate medicine and nursing students.

Design Observational, cross-sectional study using an online survey administered during the academic term in 2024.

Setting International Health Professions University in Bahrain.

Participants 279 undergraduate students from the school of medicine and school of nursing.

Primary and secondary outcome measures Physical activity levels (International Physical Activity Questionnaire-Short Form), recreational screen time (Sedentary Behaviour Questionnaire), sleep quality (Pittsburgh Sleep Quality Index) and mood (Brief Mood Introspection Scale) were measured and compared across groups, and associations between measures were assessed.

Results Participants reported high rates of not meeting physical activity recommendations (46.6%), high levels of recreational screen time (median=32 hours per week) and poor-quality sleep (63.1%). Males reported higher levels of physical activity, screen time and sleep quality. Higher sleep quality was observed for the school of medicine, the preclinical stage of study and participants living alone. Overweight and obese participants had significantly higher recreational screen time and more unpleasant and tired moods. Higher levels of screen time and lower sleep quality were associated with tired, unpleasant and negative moods, while not meeting physical activity recommendations was associated with poor sleep in addition to unpleasant, tired and negative moods.

Conclusions Physical activity levels are positively associated with mood and sleep quality in young adult healthcare students. Recreational screen time is negatively associated with mood but has no relationship with sleep quality. Intervention programmes to increase physical activity are warranted for young adults in healthcare training.

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ Reporting the lifestyle behaviours of an understudied population has contributed to expanding knowledge in the field and can help direct future research efforts.
- ⇒ Cross-sectional design does not allow identification of the directional nature of associations.
- ⇒ The generalisability of the results is limited due to the single-centre design.
- ⇒ Prospective multicentre research of the population group is recommended.

Data availability statement Study data is available on reasonable request from the corresponding author.

INTRODUCTION

The age between 18 and 29 years is recognised as a unique and distinct period of development of vital importance.¹ Several chronic disease prevention interventions have specifically addressed young adults between 18 and 29 years as a distinct group, focusing on obesity prevention and physical activity.² Studies of disease trajectory patterns have shown that low physical activity during young adulthood is associated with diabetes and hyperlipidaemia at midlife,³ while high physical activity is associated with lower incidence of cardiovascular disease events,⁴ and high television (TV) viewing at age 23 years is associated with hypertension, hyperlipidaemia and obesity at midlife.⁵ A high body mass index (BMI) at age 21 years has been considered a predictor of premature mortality.⁶

Positive correlations have been shown between physical activity and mental health within this age group.⁷ Worldwide, mental disorders are among the most common health problems,^{8 9} and the period of late

adolescence and young adulthood is when many mood disorders emerge.^{10 11} During young adulthood, identity is explored and formed with the accompanying anxieties related to relationships, entering the workforce and formulating a personal worldview.¹ Understandably, anxiety and mood disorders are highly prevalent during these years relative to older adulthood.¹²

Young adults living in the post-COVID-19 pandemic are exposed to environments where education, entertainment, instruction, socialising and just about every aspect of human existence are delivered to individuals via smartphones. Screen exposure has become an integral part of life, and young adults must embrace the smartphone culture to succeed in relationships, careers, education and training.^{13 14}

Depression is common among young adults, who inevitably have more screen exposure than older adults. This period coincides with a crucial psychological development phase. Studies have suggested that excessive screen exposure may influence mental health, such as anxiety disorders,¹⁵ depression^{16 17} and problems with sleep.¹⁸ Associations between depression and obesity are well established.¹⁹

Screen time has been considered an indication of sedentary behaviour.¹⁷ Studies have shown that sedentary behaviour associated with screen time is linked to an increased risk for cardiovascular disease,^{20 21} obesity,^{22 23} diabetes²⁴ and increased risk of death.²⁵ A positive association between sedentary behaviour and obesity is well established. Mechanistically, this may be explained by the associated co-activities of snacking behaviours and other poor dietary habits linked with excessive recreational screen time and sedentary behaviours.^{26 27}

Young adulthood is a period of occupational and educational choices where many embark on tertiary education. University, per se, has its associated stressors. These can be disproportionately higher among trainee doctors and nurses.^{28 29} Sleep disturbances are common among medical students,³⁰ and their prevalence is higher than that of non-medical students and the general population.³⁰ Recreational screen time is high among medical students, and this has been associated with poor mental health.³⁰

Understanding the factors that influence the well-being of young adult healthcare students, particularly those in high-pressure academic programmes like medicine and nursing, is of great importance as they must be physically and mentally well to provide optimal care for others. These students often experience significant stress due to their rigorous coursework and clinical duties, which can negatively impact their physical activity levels, sleep quality and overall mood. With the increasing reliance on digital devices for both academic and recreational purposes, excessive screen time has become a growing concern, potentially exacerbating these challenges. This study reports the levels of recreational screen time (using TV, computer, cell phone and tablet), physical activity, sleep quality and mood among young adult students of

medicine and nursing and any associations between these measures. By examining the relationships between these factors, this study provides valuable insights into the unique well-being needs of healthcare students, which is essential for developing targeted interventions and policies to promote healthier lifestyles and improve their overall health.

METHODS

This was an observational cross-sectional study at an international health professions university in Bahrain. Based on the university's undergraduate population size of 1300, the sample size required to reach statistical significance with a confidence level of 95% and a margin of error of 5% was calculated to be 297. Inclusion criteria for participating in the study were being 18 years of age or older, being a registered undergraduate student in either the school of nursing or the school of medicine during the 2023/2024 academic year, and giving informed consent to participate in the study. A convenience sample of the student population was recruited through a variety of channels including university email, advertising posters within the university campus, in addition to student council, class representative and academic staff announcements. The survey was opened in January 2024 during the second semester of the 2023/2024 academic year and was closed in February 2024 as this time did not overlap with any scheduled examinations. The study survey was hosted online on Microsoft Office Forms, and all recorded responses were anonymous. Participants were first provided with the participant information leaflet and confirmed electronic consent before completing the survey.

Survey instruments

The online survey had several component questionnaires covering basic anthropometrics, demographics, physical activity, recreational screen time, sleep and mood. The Short Form International Physical Activity Questionnaire (IPAQ-SF)³¹ was used to report participants' physical activity levels over the preceding week. To better facilitate accurate reporting of physical activity time in electronic format, the IPAQ-SF was modified to prompt recording of activity time in minutes rather than hours and minutes. Guidelines for Data Processing and Analysis of the IPAQ were used to categorise the participants' physical activity levels.³² Items from the Sedentary Behaviour Questionnaire (SBQ)³³ were modified to include additional screen devices (smartphone and tablet) and used to report recreational screen time. The Pittsburgh Sleep Quality Index (PSQI) was used to measure the quantity and quality of sleep over the preceding month.³⁴ The PSQI contains several subscales; however, only global PSQI scores are reported here. Finally, the Brief Mood Introspection Scale (BMIS) was used to report the current mood states of the participants.³⁵ All participants were highly proficient in the English language but not native

English speakers, so minor modifications were made to the BMIS to include definitions to ensure participants understood all adjectives included in the instrument. The BMIS: Technical and Scoring Manual (Third Edition) was used to score participants' mood based on four scales: pleasant-unpleasant, arousal-calm, positive-tired and negative-relaxed.³⁶ The four-point response scale (definitely do not feel=1, do not feel=2, slightly feel=3, definitely feel=4) and reverse scoring method were used to calculate individual scale scores. Each individual scale is a composite of items relevant to that mood scale: pleasant-unpleasant 16 items (score range 16–64), arousal-calm 12 items (score range 12–48), positive-tired seven items (score range 7–28), negative-relaxed six items (score range 6–24) with higher scores indicating a mood state closer to the first word in the scale name.

Responses with BMI values with extreme values ($<15 \text{ kg/m}^2$ or $>60 \text{ kg/m}^2$) were removed from the study. Data validation for the IPAQ and PSQI questionnaires was managed according to the data validation guidelines for these instruments.^{32–34} List-wise deletion was used to remove responses with more than 20% missing data, and mean imputation was used for responses with $<20\%$ missing data in the BMIS and SBQ items.

Statistical methods

All statistical analysis was completed using IBM Statistics SPSS V.29.³⁷ Categorical variables are reported as frequencies and percentages in parentheses. All continuous variables were tested for normality using the Kolmogorov-Smirnov test. Total metabolic equivalent of task (MET) min per week, recreational screen time (hours per week), hours asleep, PSQI score and all scales of the BMIS are presented as medians with IQR in parentheses. All between-group comparisons were completed using the Mann-Whitney U test. To allow comparisons between physical activity, recreational screen time and sleep quality, these variables were dichotomised. Physical activity was divided into not meeting recommendations (<150 min per week moderate physical activity) and meeting recommendations (≥ 150 min per week moderate physical activity).³⁸ Recreational screen time was divided into low and high categories using 4 hours per day as the threshold. PSQI scores of ≤ 5 were considered as high quality sleep and PSQI scores >5 as low quality sleep.³⁴ Where appropriate, relationships between categorical variables were examined using the Pearson χ^2 test and odds ratio (OR) in addition to the effect size, the phi coefficient, ϕ . Statistical significance of all tests was set to 0.05.

Patient and public involvement

This research study, focused on university students, was conceptualised and developed in partnership with members of the student body. Their contributions were integral to ensuring its relevance and alignment with the needs and perspectives of the student community. Students identified key issues affecting their academic

and personal experiences at university and actively participated in designing the research questions, methodology and data collection approaches used in the study. Students' input ensured that the study was sensitive to the diverse experiences of the student population and the burden that involvement in this study may place on them. Students contributed to the design of participant leaflets, consent forms and survey instruments to ensure clarity and cultural appropriateness. This study will be shared with the student body, and its results will be used to advocate for interventions and policies that support students' health and well-being.

RESULTS

In total, 295 participants consented to participate in the study and completed the survey. During data processing and validation, 16 responses were found to be invalid. They were excluded from the study due to missing or invalid data in BMI (3), IPAQ (7), SBQ screen time (0), PSQI (2) and BMIS (4) components of the survey. This left a sample of 279 eligible participants with complete, valid responses.

The study sample ($n=279$) consisted of 83 (29.7%) males and 196 (70.3%) females with a median age of 20 years and IQR from 19 to 21 years. The reported age range is typical for students enrolled in undergraduate medical and nursing programmes in this region. According to BMI category, 22 (7.9%) were underweight, 149 (53.4%) normal weight, 70 (25.1%) overweight and 38 (13.8%) obese. The majority of participants were enrolled in the school of medicine, 222 (79.6%), while 57 (20.4%) were from the school of nursing. All students were enrolled in programmes consisting of preclinical and clinical stages of study, of which 213 (76.3%) were classified as being in the preclinical stage of study and 66 (23.7%) in the clinical stage of study. The majority of participants, 195 (69.5%), were living with company, either roommates or family, as opposed to living alone. In total, 25 (9%) participants identified as being diagnosed with a mental or developmental disorder, and 37 (13.3%) reported a chronic disease diagnosis.

Physical activity was assessed and categorised based on IPAQ-SF responses: 80 (28.7%) had low, 117 (41.9%) had moderate and 82 (29.4%) had high levels of physical activity with a total of 149 (53.4%) meeting physical activity recommendations of at least 150 min of moderate intensity or 75 min vigorous intensity physical activity per week. The median recreational screen time was 32 hours per week, and the IQR spanned from 21 to 42 hours per week. The daily median number of hours asleep was 7 hours per day with an IQR of 5–8, and the median PSQI score was 7 (IQR 4–9). The medians and IQRs of participants' mood scores on the four different scales of the BMIS are reported in [table 1](#).

Normality tests confirmed that all scales (MET min per week, recreational screen time, PSQI and all parts of the BMIS) were not normally distributed. Non-parametric

Table 1 Participants' characteristics (n=279)

Demographics, recreational screen time, physical activity, sleep and mood	
Gender, n (%)	
Male	83 (29.7)
Female	196 (70.3)
Age, median (IQR)	20 (19–21)
Body mass index category, n (%)	
Underweight (BMI <19)	22 (7.9)
Normal weight (19≤BMI <25)	149 (53.4)
Overweight (25≤BMI <29)	70 (25.1)
Obese (BMI ≥31)	38 (13.6)
School, n (%)	
Medicine	222 (79.6)
Nursing	57 (20.4)
Stage of study, n (%)	
Preclinical	213 (76.3)
Clinical	66 (23.7)
Living situation, n (%)	
Alone	85 (30.5)
With company	194 (69.5)
Mental or developmental disorder, n (%)	
No	254 (91.0)
Yes	25 (9)
Chronic disease, n (%)	
No	242 (86.7)
Yes	37 (13.3)
IPAQ physical activity category, n (%)	
Low	80 (28.7)
Moderate	117 (41.9)
High	82 (29.4)
Achieved physical activity recommendations, n (%)	
No	130 (46.6)
Yes	149 (53.4)
Recreational screen time per week, median (IQR)	32 (21–42)
Recreational screen time category, n (%)	
Low (<4 hours per day)	106 (38)
High (≥4 hours per day)	173 (62)
Pittsburgh Sleep Quality Index score, median (IQR)	
Hours asleep	7 (5–8)
PSQI score	7 (4–9)
Pittsburgh Sleep Quality Index category, n (%)	
Low (PSQI >5)	176 (63.1)
High (PSQI ≤5)	103 (36.9)

Continued

Table 1 Continued

Demographics, recreational screen time, physical activity, sleep and mood	
Brief Mood Introspective Scale, median (IQR)	
Pleasant–unpleasant scale	42 (38–47)
Arousal–calm scale	32 (29–35)
Positive–tired scale	19 (16–21)
Negative–relaxed scale	15 (12–18)

tests were therefore used for all comparisons. Significant differences between males and females were identified in MET min per week, hours per week of recreational screen time, PSQI score and the arousal–calm scale of the BMIS.

Males reported higher MET min per week, 2447 (IQR 1110–3990), compared with females, 1288 (IQR 466–2691). The Pearson χ^2 test indicated a significant association between gender and physical activity categories ($\chi^2=3.877$, $p=0.049$); however, the OR was not significant. Males also reported higher median recreational screen time, 35 (IQR 27–44) hours per week, compared with females, 29 (IQR 20–42) hours per week. An association was also observed when comparing gender and recreational screen time categories ($\chi^2=8.078$, $p=0.004$), which indicated males were more than two times as likely as females to engage in ≥4 hours of recreational screen time per day (OR=2.26, 95% CI (1.279 to 3.997)), with a small-to-medium effect size ($\phi=0.17$).

Females had significantly higher median PSQI score, 7 (IQR 4–10) versus 6 (IQR 4–9), indicating they experienced lower quality sleep compared with males. However, significant gender differences in frequencies of low (PSQI >5) and high (PSQI ≤5) categories of sleep quality were not detected.

Female participants had a higher median score on the arousal–calm scale in the BMIS, 32 (IQR 30–35) versus 31 (IQR 28–43), indicating they were less calm and more aroused compared with male participants.

Only one scale was significantly different between the school of medicine and the school of nursing. Participants enrolled in the school of nursing had a higher median PSQI score, 8 (IQR 6–11) versus 7 (IQR 4–9). Similar results were seen for groups based on stage of study. Preclinical participants differed from clinical participants in PSQI score, having a median of 8 (IQR 6–10) compared with the median for preclinical participants of 6 (IQR 4–9). Participants in their clinical stages were found to be two times as likely to have low quality sleep compared with preclinical participants ($\chi^2=5.964$, $p=0.015$), (OR=2.15, 95% CI (1.155 to 4.032)), $\phi=0.15$. The observed sleep quality associations across stages of study and schools are deemed to be linked. This may be due to the relative proportions of participants in the different stages in our sample from each school. The proportion of clinical students in the school of nursing sample was biased towards the clinical stage (86%), while

the school of medicine was heavily biased towards the preclinical stage (93%).

Participants' living situation was observed to show differences in both the physical activity and sleep domains. Participants living alone had higher levels of MET min per week, 1760 (IQR 900–3813) and better sleep quality, 6 (IQR 4–9), compared with those who lived with company, respectively 1386 (IQR 456–2820) and 7 (IQR 5–10). Participants living with company were two times as likely to report poor quality sleep compared with those living alone ($\chi^2=5.398$, $p=0.020$), (OR=1.85, 95% CI (1.098 to 3.115)), with a small-to-medium effect size ($\phi=0.14$). This observed association may also be linked to the stage of study association with sleep, as 96.5% of those living alone were in the preclinical stage of study, compared with 67.5% of participants who were living with company who were in the preclinical stage.

Participants who declared a diagnosis of a mental/developmental disorder or a chronic disease both showed significant differences in three of the four BMIS scales (pleasant–unpleasant, positive–tired, negative–relaxed) compared with those who did not report such a diagnosis. Similar patterns were seen in group differences for either diagnosis, typified with a diagnosis being associated with lower scores for both the pleasant–unpleasant scale and positive–tired scale and higher scores observed in the negative–relaxed scale of the BMIS. Each observed difference suggests that those who reported a diagnosis had less positive or more negative mood states compared with those without a diagnosis. Additionally, participants with a chronic disease diagnosis reported a significantly higher median PSQI score, 9 (IQR 7–12), compared with those without a chronic disease, 7 (IQR 4–9). Participants with a chronic disease diagnosis were more than three times as likely to have reported low quality sleep compared with those who did not ($\chi^2=7.850$, $p=0.005$), (OR=3.46, 95% CI (1.390 to 8.597)), with a small-to-medium effect size ($\phi=0.17$).

Underweight/normal weight and overweight/obese groups showed significant differences in recreational screen time, pleasant–unpleasant scale and positive–tired scale. The underweight/normal weight group reported significantly less screen time, 28 (IQR 20–37) hours per week, compared with the overweight/obese group, 37 (25–45) hours per week. Underweight/normal weight participants were two times as likely to report <4 hours per day of recreational screen time compared with overweight and obese participants ($\chi^2=4.137$, $p=0.042$), (OR=1.69, 95% CI (1.017 to 2.817)), $\phi=0.12$. Higher median scores on the pleasant–unpleasant and positive–tired scales of the BMIS were detected for the underweight/normal weight group, table 2, compared with the overweight/obese group.

Participants who did not meet physical activity recommendations had significantly different median PSQI scores, 7 (IQR 5–10) compared with those who did meet recommendations, 6 (IQR 4–9), indicating participants with lower levels of physical activity experienced

lower sleep quality; however, no association was observed between achieving physical activity recommendations and sleep quality categories. Participants who achieved physical activity recommendations also showed significantly higher median scores on the pleasant–unpleasant and positive–tired scales. Relatively higher scores on the negative–relaxed scale were reported by participants who did not achieve recommendations.

Participants who reported less than 4 hours per day of recreational screen time showed significant differences in the pleasant–unpleasant, positive–tired and negative–relaxed scales of the BMIS compared with those who had more than 4 hours per day. The group with lower levels of recreational screen time had higher scores in both the pleasant–unpleasant and positive–tired scales and lower scores in the negative–relaxed scale, table 2.

Differences in the median scores on three of the four BMIS scales (pleasant–unpleasant, positive–tired, negative–relaxed) were detected between low-quality and high-quality sleep groups. The observed trend paralleled those of achieving physical activity recommendations and recreational screen time levels, with higher scores in both the pleasant–unpleasant and positive–tired scales and lower scores in the negative–relaxed scale.

DISCUSSION

Our study identified an association between high physical energy expenditure, male gender and living alone. Low physical activity levels were associated with poor sleep quality scores, tired and unpleasant mood. There was no association between physical activity levels and recreational screen time exposure. There were no differences in physical activity levels between nursing and medical students or preclinical or clinical students. Previous studies have also shown male students to have higher physical activity levels as compared with female students.^{7 39 40} We reported high physical activity among participants who lived alone, which reflected the findings of a study of American students in private universities, which found higher physical activity levels for students who lived without a parent or guardian.⁴¹ Our findings of an association between physical activity and sleep quality are consistent with a meta-analysis showing that higher levels of physical activity are associated with higher quality sleep in college students⁴² and older adults.⁴³ Positive correlations have been found between physical activity and mental health,⁷ which reflects our findings concerning low physical activity with tired and unpleasant mood. More specifically, a study of university students in Ireland reported that students who met physical activity recommendations were happier than those who did not.⁴⁴ Blake *et al* had previously reported significant differences in physical activity levels between medicine and nursing students. They revealed several barriers to achieving recommended physical activity levels that nursing students were confronted with.⁴⁵ While our study found a slightly higher proportion (46.6%) of total participants did not

Table 2 Between-group comparisons of median (IQR) measures of physical activity, screen time, sleep and mood

	MET min per week	Recreational screen time (hours per week)	PSQI score	Pleasant-unpleasant scale	Arousal-calm scale	Positive-tired scale	Negative-relaxed scale
Gender							
Male (n=83)	2447 (1110–3990)*	35 (27–44)*	6 (4–9)*	42 (38–48)	31 (28–34)*	19 (16–21)	14 (12–17)
Female (n=196)	1288 (466–2691)	29 (20–42)	7 (4–10)	42 (37–47)	32 (30–35)	19 (16–21)	15 (13–18)
School							
Medicine (n=222)	1535 (650–3164)	32 (21–42)	7 (4–9)*	42 (38–48)	31 (29–34)	19 (16–21)	15 (12–18)
Nursing (n=57)	1533 (282–2781)	34 (23–43)	8 (6–11)	42 (36–47)	33 (30–36)	19 (17–21)	16 (13–18)
Stage of study							
Preclinical (n=213)	1528 (622–3215)	30 (21–42)	6 (4–9)*	43 (38–48)	32 (29–35)	19 (17–21)	14 (12–18)
Clinical (n=66)	1733 (297–2809)	35 (23–43)	8 (6–10)	42 (36–47)	32 (29–35)	19 (16–21)	16 (13–18)
Living situation							
Alone (n=85)	1760 (900–3813)*	34 (21–43)	6 (4–9)*	43 (38–49)	31 (29–34)	19 (17–22)	14 (12–17)
With company (n=194)	1386 (456–2820)	30 (21–42)	7 (5–10)	42 (37–47)	32 (29–35)	19 (16–21)	15 (12–18)
Mental/developmental disorder							
Yes (n=25)	1386 (405–2969)	32 (18–42)	9 (6–13)	35 (33–42)*	32 (28–35)	17 (13–19)*	17 (14–20)*
No (n=254)	1618 (518–3077)	32 (21–42)	7 (4–9)	43 (38–48)	32 (29–35)	19 (17–21)	15 (12–17)
Chronic disease							
Yes (n=37)	1584 (488–2819)	33 (18–47)	9 (7–12) *	40 (35–44)*	33 (29–35)	18 (16–20)*	17 (14–20)*
No (n=242)	1531 (540–3110)	32 (21–42)	7 (4–9)	43 (38–48)	31 (29–35)	19 (16–21)	15 (12–17)
BMI category							
Underweight/normal weight (n=171)	1533 (617–3052)	28 (20–37)*	7 (4–9)	44 (39–48)*	32 (29–35)	19 (17–21)*	15 (12–17)
Overweight/obese (n=108)	1573 (462–3175)	37 (25–45)	8 (5–10)	41 (35–46)	31 (28–35)	19 (16–21)	16 (13–19)
Achieved physical activity recommendations							
No (n=130)	–	32 (21–42)	7 (5–10)*	41 (35–46)*	31 (28–34)	18 (16–21)*	16 (13–18)*
Yes (n=149)	–	32 (21–42)	6 (4–9)	44 (39.5–49)	32 (30–35)	20 (17–21)	14 (12–17)
Recreational screen time category							
Low (<4 hours per day) (n=106)	1618 (476–2973)	–	7 (4–10)	44 (40–49)*	32 (30–34)	20 (18–21)*	14 (12–17)*
High (≥4 hours per day) (n=173)	1528 (570–3149)	–	7 (4–9)	42 (36–47)	32 (29–35)	19 (16–21)	16 (13–18)
Sleep quality category							
Low (PSQI >5) (n=176)	1386 (484–2991)	32 (22–42)	–	40 (35–45)*	32 (29–35)	19 (16–20)*	16 (13–19)*
High (PSQI ≤5) (n=103)	1674 (655–3195)	32 (21–41)	–	47 (42–50)	31 (29–34)	21 (18–22)	13 (11–16)

*Statistically significant difference in median values between groups (p<0.05). BMI, body mass index; MET, metabolic equivalent of task; PSQI, Pittsburgh Sleep Quality Index.

meet physical activity recommendations as compared with 42.9% reported by Blake *et al*, no difference in physical activity levels between nursing and medicine groups of students was detected in our study.⁴⁵ While our study reports physical activity, the perceived motives were not investigated. A study conducted on medical field students across several institutions and countries in the Western Balkans reported students' perceived barriers and motivation for physical activity; important motivations include feeling better, reduction of stress and aesthetics, while faculty obligations was the notable barrier reported.⁴⁶ These results are aligned with a multinational systematic review, which demonstrated that physical activity is associated with improved quality of life and reduced burnout among medical students.⁴⁷

High recreational screen time was associated with being overweight, male gender, tiredness, unpleasantness and negative mood. There was no association between screen time and feeling calm or aroused. Similarly, there was no association between screen time exposure and the academic course of study (nursing or medicine, preclinical or clinical). A study designed to describe changes in recreational screen time behaviour patterns among young American adults during the COVID-19 pandemic reported a weekly average of 25.9±11.9 hours during 2018 and 28.5±11.6 hours in 2020.⁴⁸ Our study reported average hours higher than those surveyed during the pandemic period. An association between screen time exposure and obesity is well established in the literature,^{49 50} which was reflected in our data. With regards to gender differences in screen time, a study from Serbia showed female medical students reported higher scores of social media addiction as compared with males, but with no differences in general smartphone addiction between the genders.⁵¹ Our study did not specifically examine social media usage; however, based on screen time alone, our results seem incongruent with these findings, as we found significantly more screen time exposure among male students. Another study reported more screen time exposure for male than female respondents, which was our finding.⁵² Generally, studies suggest that total recreational screen time hours per week are higher for males than females and that males exhibit a tendency towards gaming, while females tend towards social media.⁵³ The same Serbian survey found smartphone addiction to be associated with poor sleep quality, anxiety and depression. The authors found that more than 4 hours of daily screen exposure was associated with increased levels of depression.⁵¹ However, there is a disparity in the literature regarding screen time exposure as related to mood states and young adults' well-being; this was reflected in our observations across the four domains of mood studied. A Norwegian study of students showed a strong inverse correlation between screen time and sleep quality;⁵² however, we found no discernible relationship between screen time and sleep.

Poor sleep quality was associated with female gender, living with company, studying nursing and being in the clinical years of the course. Gender differences in sleep

quality have been previously reported in young adults.⁵⁴ Medical students have been reported to average 6.5 hours of sleep per night, which is below the recommended 7–9 hours.⁵⁵ Much has been made about the high prevalence of sleep problems among medical students.^{55–57} However, our study suggests that nursing students' sleep quality is a more significant issue. A study from two Mediterranean countries shows this difference in sleep quality between nursing and medical students.⁵⁸ Longitudinal studies have shown that this issue of poor sleep quality follows student nurses into their professional careers long after graduation,⁵⁹ with further declines in sleep quality for the first 3 years of clinical practice.⁵⁹

Given that these health behaviour issues can persist from university into professional practice, the implementation of strategies and interventions to mitigate such behaviours is of great importance. A meta-analysis of randomised controlled trials of interventions to increase physical activity in university students found that a wide range of approaches, including online theory-based programmes, behavioural monitoring, gamification, activity trackers and credited course interventions, were effective in increasing total physical activity.⁶⁰ A recent paper described an effective intervention to enhance nursing students' well-being through an 8-week digital detox programme and reported significant reductions in scores on the Social Media Addiction Scale. The programme incorporated digital hygiene education, self-reflection exercises, goal setting, screen time management strategies, mental wellness modules, physical activity promotion and hobby-based alternatives.⁶¹ These results highlight the potential of structured digital detox initiatives to support healthier technology use, improve mood and promote overall well-being in healthcare students.

Limitations

This study's cross-sectional design offers only a 'snapshot' of participants' responses, limiting the ability to determine causal relationships. Self-reported instruments, while well-established, introduce recall and social desirability biases, especially in this health-literate sample group. Subjective physical activity, sleep and screen time assessments may overestimate or underestimate behaviours.^{62–66} The IPAQ has excellent reliability, but only moderate correlation with objective measures of physical activity.⁶⁷ Studies of physical activity over-reported the proportion of participants meeting recommendations by subjective measures compared with objective measures of physical activity.⁶⁸ While good reliability and validity are reported for the PSQI,⁶⁹ it does not equate to the gold standard; polysomnography. Consumer sleep-tracking devices exist but lack sufficient validation.⁷⁰ Self-reported screen time validity is low, yet subjective measures persist due to practical challenges in direct observation across multiple devices.⁷¹ Minor modifications to the BMIS and SBQ may have impacted validity. The SBQ does not distinguish between passive media consumption and interactive media

engagement with social media, and hence may mask distinct interactions between sleep and mood.

The convenience sampling method potentially introduced selection bias. Students with particular interest or concern in the study's domains (eg, sleep and exercise) may be over-represented. The study also has an over-representation of preclinical and medicine students. The sample includes a much higher proportion of medicine versus nursing students and preclinical versus clinical students. These imbalances potentially compromise representativeness and may distort associations observed between sleep quality and school or stage of study. Non-response bias may have influenced the results of our research.⁷² Students who acknowledge their own lack of sleep or physical exercise may be less likely to respond to a request to participate in the survey.

Most participants were international students from countries in the Middle East and North Africa, as well as North America and the Indian subcontinent. Our survey did not differentiate local versus international students, potentially overlooking important demographic influences.

The study's single-centre and medical university-specific context limits generalisability. Findings may not be extrapolated to other healthcare disciplines (eg, allied health, dentistry), different cultural or educational settings or to broader young adult populations. The unique nature of medical education programmes and the high proportion of international students may further restrict extrapolation.

Finally, important confounders—including academic workload, stress levels, caffeine and alcohol intake—were not measured or inadequately controlled for, potentially influencing interpretations of relationships between sleep, mood and screen time.

Conclusions

A substantial proportion of medicine and nursing students are not achieving the recommended levels of physical activity. This low degree of engagement in exercise may have long-term consequences for the healthcare profession based on the predicted chronic disease trajectory from young to middle-aged adults. A lack of physical activity is associated with low mood and poor sleep quality among young adult healthcare students. In the short term, there are also consequences for this low physical activity as healthcare professionals are regarded as role models for health, and their physical activity behaviours may be scrutinised by society. These results suggest multi-centre prospective studies are warranted for young adults in healthcare training to determine the extent and direction of the relationships, in addition to intervention programmes to mitigate the issues related to physical activity, screen time, sleep and mood in this population.

Contributors AKA made substantial contributions to study conception, methodology, data curation, data analysis and drafting of the manuscript. FNS made substantial contributions to study conception, methodology, data curation and analysis, and drafting of the manuscript. SF made substantial contributions to

data curation, data analysis and drafting of the manuscript. DG made substantial contributions to study conception, methodology, data curation, data analysis and drafting of the manuscript. DG is the guarantor.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient and public involvement Patients and/or the public were involved in the design, or conduct, or reporting, or dissemination plans of this research. Refer to the Methods section for further details.

Patient consent for publication Not applicable.

Ethics approval Research ethics approval (REC/2023/212/05-Dec-2023) was granted by the RCSI Bahrain Ethics Committee on 5 December 2023. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request.

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